

Logarithms and Their Properties



Evaluating logarithms

1 Evaluate each logarithm exactly:

a $\log_2 32$

b $\log_3 \frac{1}{9}$

c $\log 1000$

d $\ln e^7$

2 Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.

3 State the domain of $f(x) = \log(x - 4)$ and the inverse function of $g(x) = 2^x$.

Properties of logarithms

4 Expand fully using properties of logarithms: $\ln \left(\frac{x^3 \sqrt{y}}{z^2} \right)$.

5 Write as a single logarithm: $2\log x + 3\log y - \log z$.

6 Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:

a $\log_b 6$

b $\log_b 8$

c $\log_b 1.5$